



BCP 14 breakout session: adopting BCP 14 requirement levels in CF

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NATURAL ENVIRONMENT RESEARCH COUNCIL



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The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

- **BCPs** are stable identifiers for Best Current Practices
- We want to apply these to the full CF document, see [Issue #258](#) (started 2023)
- [BCP 14](#) consists of two related standards:
 - **RFC 2119**: Defines terms like MUST, SHALL, REQUIRED, SHOULD, RECOMMENDED, MAY, and OPTIONAL to establish strict technical requirements
 - **RFC 8174**: Updates RFC 2119 by clarifying that these words only carry their special requirement definitions if written in all-uppercase (e.g. MUST)



Illustrating the concept

- **should** ≠ **SHOULD**, conveying different intention and context:
 - From general conversation: “You **SHOULD** back up any important files regularly.” [if give it BCP intention], vs. “The train **should** arrive on time.” [expectation, not a recommendation], vs. “It’s raining so you **should** bring an umbrella” [advice, not formal intention]
 - From CF: “The conventions **should** be practicable for both producers and users of data.”, “The metadata **should** be both easily readable by humans and easily parsable by programs.”, but “The units attribute **SHOULD** be present for all variables that contain dimensional quantities.”
- Note: one may not use the exact words, but regardless have the same meaning, one goal of the hackathon could be to identify the underlying requirement level, rather than mechanically replacing words. Example: “**It is suggested that** a zero offset be stated explicitly to avoid confusion...” is a RECOMMENDED case without using the word itself.



For the breakout/hackathon session

- **This topic was a breakout session at the 2024 CF Workshop** and good progress was made, see https://docs.google.com/document/d/1VknWozJM3vkUeoJKQHXLmroXPwZtd2EO8_jgGIIID-kw/edit?tab=t.0 (GDoc notes) and <https://github.com/cf-convention/cf-conventions/issues/546> (follow-on Issue for implementation, now closed as “not planned”). Daniel Lee’s summary of status:
 - *We've decided that we want to use BCP 14 unmodified, and that the "EBCP" words are to help us ensure that we're actually saying what we mean. The rendering is still an open issue, although I believe we can address that a bit further down the road.*
 - *“The rendering”: some thought all upper case is “too shouty” so TBC whether to adopt that i.e. RFC 8174 way to indicate RFC 2119*
- **Lars has done excellent work to analyse the document to e.g. provide counts of the BCP keywords** so much work is required for a given one in any conversion
- **BCP adoption can't go in the document piecemeal**, so aim to get a Pull Request fully implementing it prepared to be merged and adopted in time for the release next year (v1.15), ideally
- **First and foremost, we need to agree a strategy for getting BCP in.** Suggestion to get the ball rolling!
 - Identity all statements with the BCP keywords or equivalent wording (Lars’ work helps)
 - Big table of all of these with checklist to tick off thoughts of BCP case or not?
 - In the age of AI tools/agents, could we use these to make a first pass and then have N (2?) humans confirm if they agree - could speed up the process?

